Medalytics Process Book

Dive into the Olympic Games

A project by
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For the
COM-480 - Data visualization
course

Introduction

Medalytics is an interactive data visualization platform that explores the rich and evolving history of the Olympic Games. From the first modern Games in 1896 to the global multi-sport spectacle it is today, the Olympics offer a unique lens on international representation, gender dynamics, sports diversity, and national success.

Our project seeks to reveal meaningful insights through engaging, animated, and user-friendly visualizations. Each section of the website is designed to guide users on a journey, highlighting major trends, anomalies, and hidden patterns in Olympic history.

This process book documents the development of the project: from ideation and data preprocessing to visualization design and technical challenges. It reflects both the creative and analytical decisions that shaped our final product.

Data Preprocessing

Using a variety of datasets — including athlete records, event results, and country-level medal statistics, we've curated a scrollytelling experience that merges storytelling with interactivity. Whether you're a casual sports fan or a data enthusiast, Medalytics offers a new way to explore the Games.

For this project, we used multiple datasets from a GitHub repository curated by MIT graduate *Keith Galli*, originally sourced from **Olympedia.org**. These datasets cover both summer and winter Olympic Games from **1896 to 2022**, and include cleaned versions with parsed dates, standardized fields, and removed formatting noise.

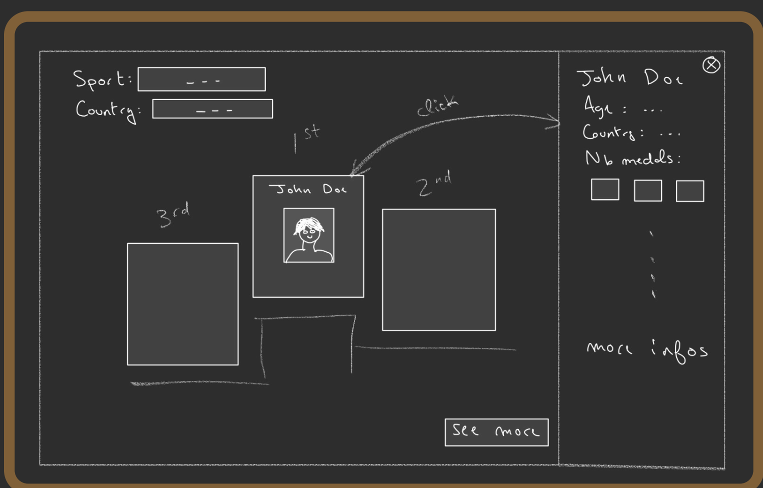
Although the data was relatively well-prepared, we still encountered inconsistencies — such as mixed naming conventions (e.g., "USA" vs "United States") and column name discrepancies (e.g., noc vs NOC). To address these issues, we implemented a comprehensive data cleaning pipeline that standardized country names, unified column naming conventions, and ensured consistent data formats across all datasets.

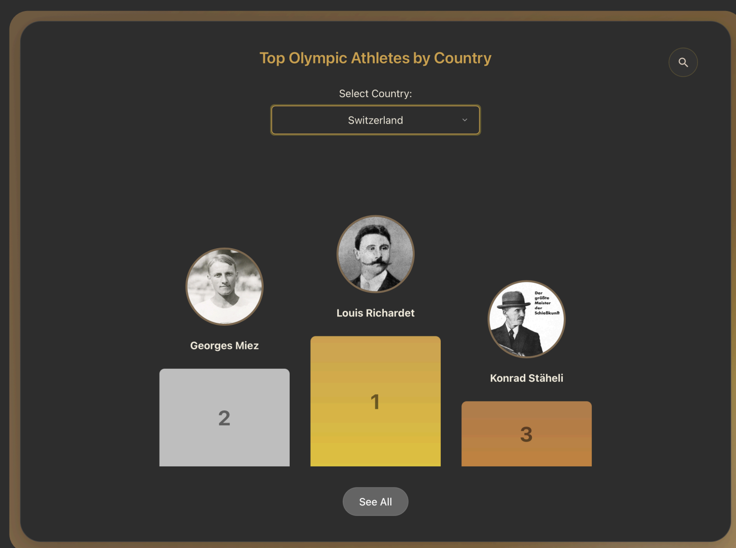
We used **PapaParse**, a powerful CSV parsing library, to efficiently process and transform our data. This allowed us to handle large datasets with ease while maintaining data integrity. We also integrated athlete biographical data, event results, and population statistics to enrich the context of each visualization. Additional preprocessing included merging tables, transforming formats into JSON for D3 visualizations, and tagging athletes with continent metadata using pycountry-convert.

Visualizations

Olympic podium

The Olympic Podium creates an engaging way to explore Olympic athletes **by country**, inspired by the iconic medal ceremony stage. Users can select any country to see its top three athletes ranked by their medal achievements, with each athlete displayed on a podium block styled in Olympic medal colors (gold, silver, and bronze). The visualization brings the excitement of Olympic victory ceremonies to life through smooth animations and dynamic interactions.



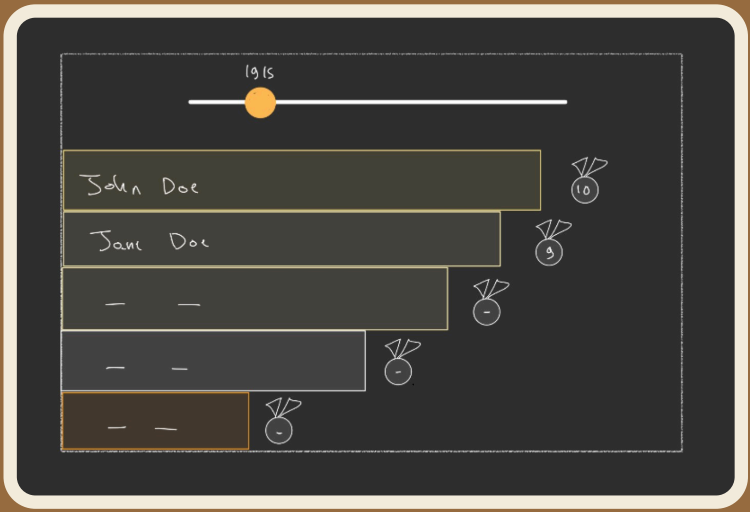


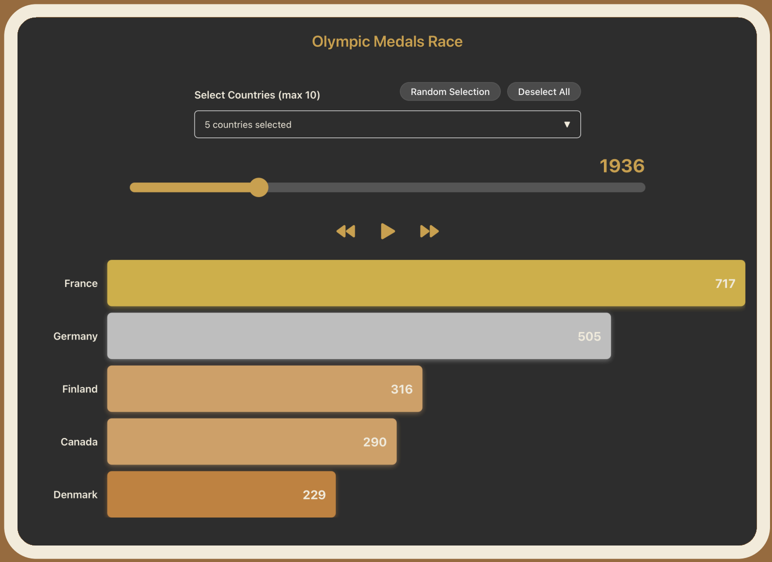
The visualization offers multiple **layers of exploration**: clicking on an athlete reveals their detailed profile, complete medal history, and chronological list of Olympic achievements. A "**See All**" feature expands to show a comprehensive list of athletes from the selected country, while a search function allows users to find specific athletes across all nations. This multi-layered approach ensures that users can easily navigate from a high-level overview to detailed individual achievements.

The implementation focuses on smooth transitions between views and intuitive visual hierarchy, using **medal colors and podium positioning** to create an immediate understanding of athlete rankings. Each athlete's profile includes their complete Olympic journey, from their first appearance to their most recent achievements, providing context to their medal counts. The visualization's success lies in its ability to transform raw medal data into an interactive experience that celebrates Olympic achievement while maintaining easy access to detailed information.

Medal race chart

The Medal Race Chart creates a dynamic visualization that shows how countries compete for Olympic medals **throughout history**. Users can select **up to 10 countries** to compare their cumulative medal counts, watching as the rankings evolve over time through an animated bar chart race.





The implementation features an interactive timeline with **play/pause controls**, allowing users to watch the medal race unfold year by year or jump to specific Olympic Games. The bars are color-coded based on medal counts, creating a visual hierarchy that makes it easy to track each country's performance. A **dropdown menu** enables country selection, with options to randomly select countries or start fresh.

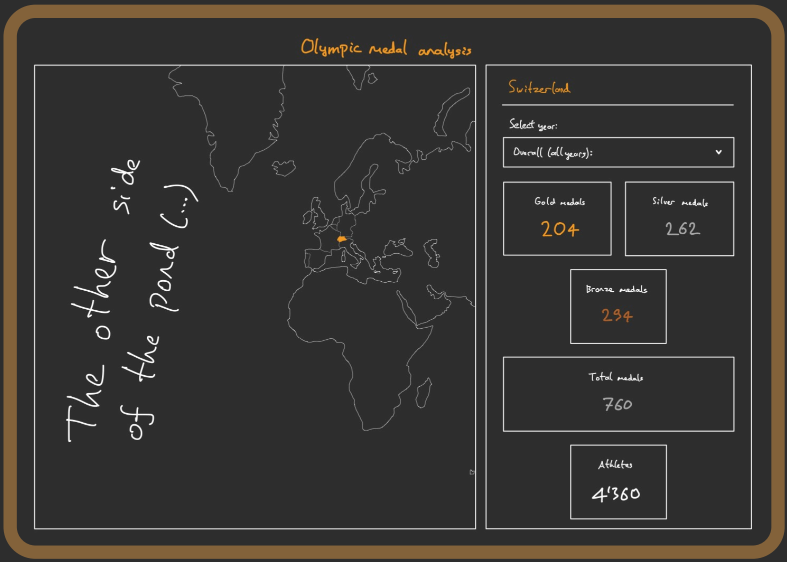
What makes this visualization particularly effective is its ability to show both the progression of Olympic success and the competitive dynamics between nations. The cumulative medal count approach reveals **long-term trends and power shifts** in Olympic history, while the animation brings the data to life in a way that static charts cannot. The implementation includes smooth transitions between years and responsive bar heights that adjust based on the number of selected countries.

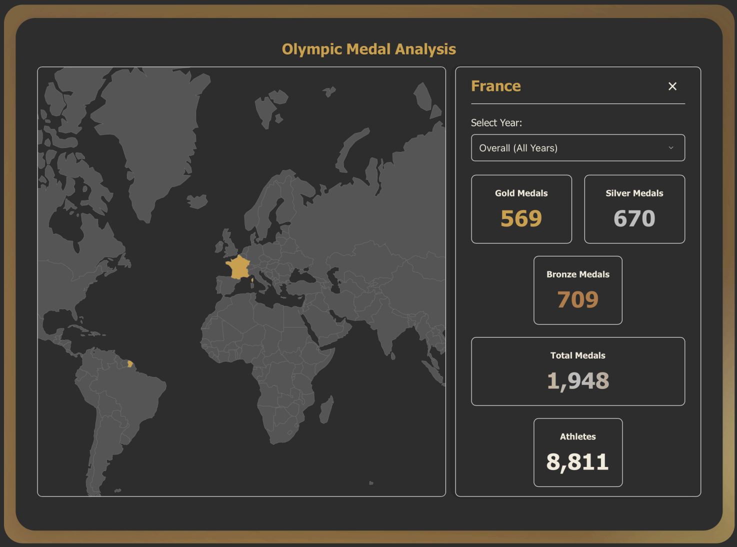
The visualization's success lies in its combination of historical data presentation and interactive features, allowing users to explore Olympic history through the lens of national achievement. The timeline controls and country selection options make it easy to focus on specific periods or compare particular nations, while the animated bars provide an engaging way to understand the evolution of Olympic success over time.

World map

The World Map creates an interactive geographic visualization of Olympic participation and success across the globe. Users can explore medal counts and athlete participation for each country, with the ability to zoom, pan, and click on countries to view **detailed statistics**.

The implementation uses **D3.js** to create a responsive world map that highlights countries based on their Olympic achievements. When a country is selected, a side panel appears showing detailed medal statistics, including gold, silver, and bronze counts, total medals, and the number of participating athletes. The map also includes a **year selector**, allowing users to view data from specific Olympic Games or see overall historical totals.





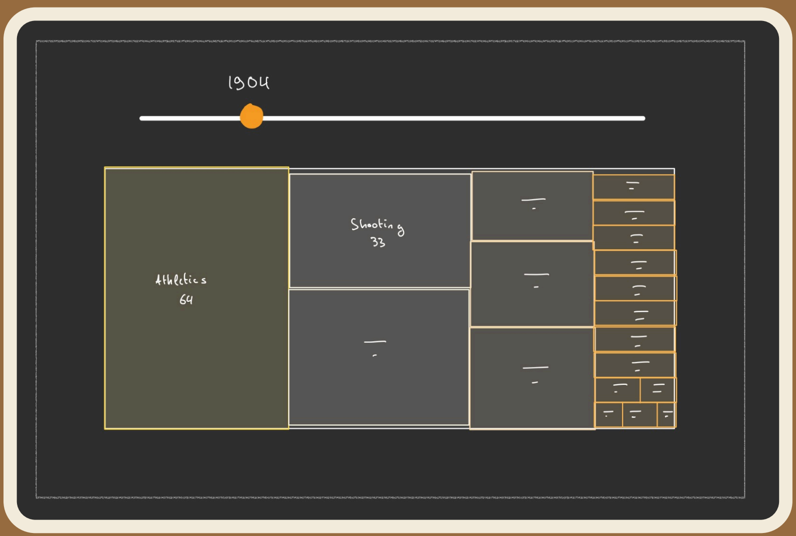
What makes this visualization particularly effective is its ability to provide both a **global perspective and detailed country-specific information**. The color intensity of each country on the map corresponds to its Olympic success, creating an immediate visual understanding of Olympic powerhouses and emerging nations. The interactive elements, including zoom and pan capabilities, make it easy to explore regions of interest, while the detailed statistics panel provides context to the visual representation.

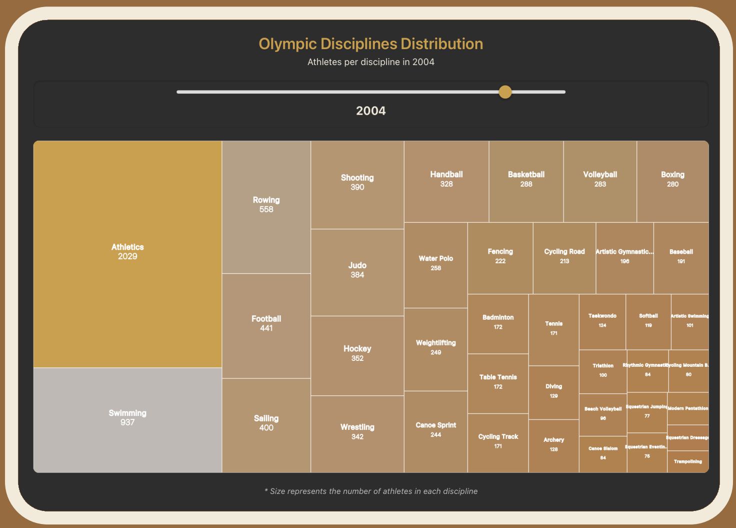
The visualization's success lies in its combination of geographic data presentation and detailed statistical information, allowing users to understand both the global distribution of Olympic success and the specific achievements of individual nations. The implementation includes smooth transitions between different views and years, making it easy to track changes in Olympic performance over time.

Sports Treemap

The Sports Treemap presents a hierarchical view of Olympic sports participation through an interactive mosaic of disciplines. Each tile in the treemap represents a sport, with its **size reflecting the number of participating athletes** and its **color indicating participation levels** using an Olympic-themed gradient. This visualization offers a unique perspective on how different sports have evolved and gained prominence throughout Olympic history.

The implementation uses **Recharts' Treemap component** to create a dynamic visualization where sports are organized in a space-filling layout. A custom content renderer ensures that sport names and athlete counts remain visible regardless of tile size, while a year slider allows users to explore how participation patterns have evolved across different Olympic Games. The visualization's responsive design adapts to different screen sizes, maintaining readability and usability across various devices.





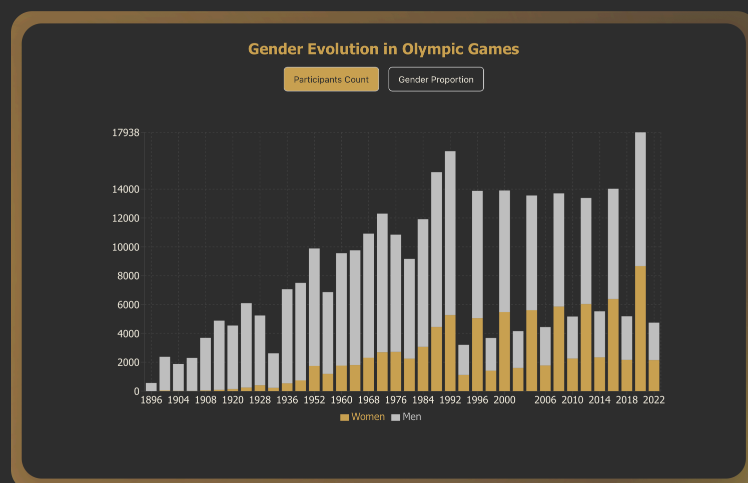
What makes this visualization effective is its ability to reveal **participation trends** and patterns that might be hidden in traditional charts. The hierarchical layout naturally groups related sports together, while the temporal navigation enables users to witness the **evolution of sports popularity** throughout Olympic history.

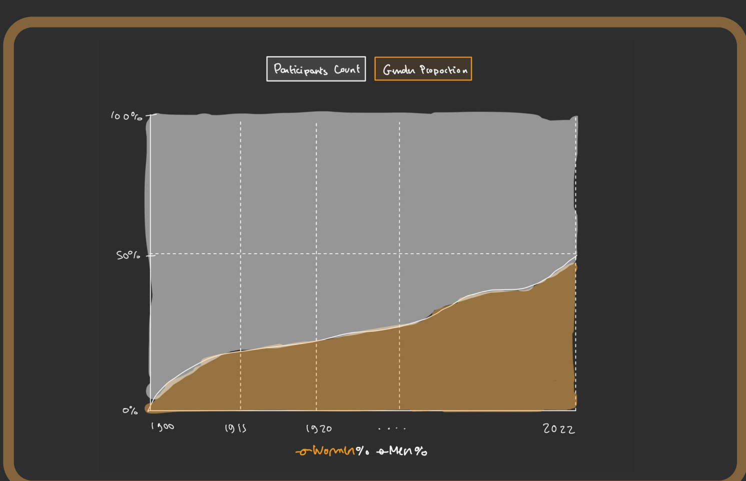
The visualization's adaptive design ensures that even smaller sports remain accessible, with tooltips providing detailed statistics on hover. The success of this visualization lies in its ability to transform complex participation data into an intuitive experience. Users can quickly identify popular sports, track changes in participation over time, and understand the relative importance of different disciplines in the Olympic landscape.

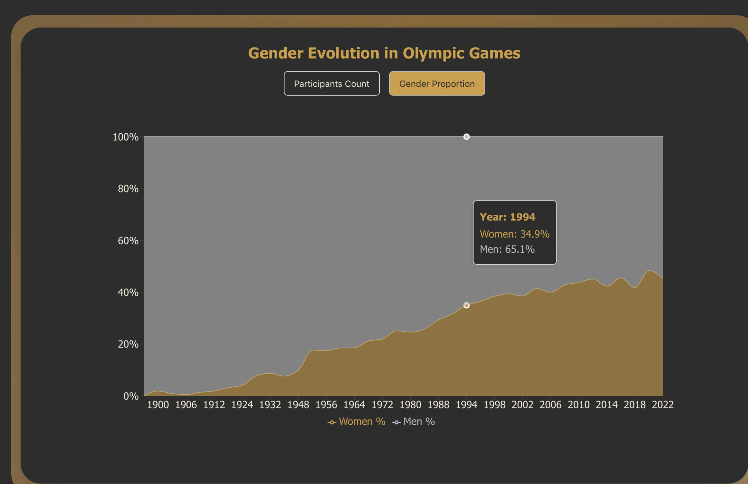
Gender chart

The Gender Evolution Chart tracks the **progression of gender equality** in Olympic participation over time, offering two distinct visualization modes to analyze this important aspect of Olympic history. Users can switch between viewing **absolute participant numbers** and **gender proportions**, providing different perspectives on the evolution of gender representation in the Games.

The implementation uses **Recharts' BarChart and ComposedChart components** to create two interactive visualizations. The first mode displays stacked bars showing the total number of male and female participants each year, while the second mode presents an area chart showing the percentage distribution between genders. The visualization features Olympic-themed colors, with gold representing female participants and silver representing male participants.







Conclusion & Considerations

Building Medalytics has been both a technical and creative journey through the history of the Olympic Games. Our goal was to go beyond static tables and charts to create an engaging, exploratory experience that reveals how the Games and the world have evolved.

Throughout development, we faced several challenges:

- **Data cleaning** required extensive attention, especially reconciling naming inconsistencies across countries and years using PapaParse and custom cleaning scripts
- Design constraints pushed us to make thoughtful choices around layout, responsiveness, and **dark/light mode** support
- **Performance optimization** was key for interactivity, particularly in animations and large data tables
- Integration of **multiple visualization libraries** (Recharts, D3.js) required careful coordination to maintain consistent styling and behavior

Despite these challenges, we are proud of the final product. The Medalytics platform is more than a collection of charts - it's a storytelling environment that encourages users to discover insights on their own terms. The combination of scrollytelling, interactive visualizations, and responsive design creates an immersive experience that makes Olympic history accessible and engaging.

Team Contributions:

- *Albert* focused on UI design, interactivity logic, and Figma prototyping
- *Daniel* led data preprocessing, animation logic, and React integration
- *Hugo* handled visualizations, component styling, and responsive layout optimization

This project taught us the importance of aligning narrative intent with technical execution — and how good design can turn complex data into accessible stories. The use of modern web technologies and visualization libraries allowed us to create a platform that is both technically sophisticated and user-friendly.